Name Patterns in the Succession of Rulers of the Holy Roman Empire

A reproducible comparison of the periods 911-1313 and 1314-1806, together with the 113-year progression of the predecessors of Conrad I-IV

Data basis	Event analysis	Joint test
64 ruler positions	52 predecessor-successor assignments	1,000,000 permutations

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All statistical statements refer to the permutation null models explicitly described in the article.

Abstract and principal result

This article examines two connected regularities in the sequence of kings, co-kings, and anti-kings of the East Frankish kingdom and the Holy Roman Empire.

1. From 911 to 1313, the sequence of names forms a largely closed chain pattern. Four blocks begin with **Conrad, Henry** and end with **Henry**. The number of names between the beginning and end of the blocks increases from 3 to 4, 5, and 6.
2. The years in which the respective predecessors of Conrad I, II, III, and IV died are 911, 1024, 1137, and 1250. They form an exact arithmetic progression with three intervals of 113 years.

The main analysis compares the historically delimited period 911-1313 with the following period 1314-1806. The search is not restricted to the already identified Conrad-Henry motif. Under the implemented rules it may select arbitrary names, either of two pattern families, the most favorable subsection, and either constant or unit-increasing gaps. Every choice available for the historical data is available again in every random permutation.

The principal results are:

Test	Result
Complete pattern search, 911-1313	**p = 0.0160**
Pattern search, 1314-1806	no admissible chain of at least three blocks
Direct comparison, “early stronger than late”	**p = 0.0009**
113-year progression, Conrad and years fixed	exact **p = 0.00000369**
113-year progression, name free and globally corrected	exact **p = 0.00015514**
Name chain and 113-year progression jointly	Monte Carlo **p = 0.000097**

The joint value is based on one million random permutations. Ninety-six of them produced both a name chain at least as strong as the observed chain and an allocation of the four 113-year events at least as strong as the observed allocation. The 95% Wilson interval for the Monte Carlo hit rate is approximately **0.000079-0.000117**.

Under the specified permutation null models, these results are clearly significant. They do not prove a particular historical cause or establish that the succession was “non-random” in a general sense. They show that the specifically defined ordering features rarely occur with equal or greater strength when the observed names are randomly reassigned.

1. Starting point and research question

1.1 The observed name chain

The starting point is the succession between Conrad I and Henry VII. With given names standardized, it can be divided into four blocks:

```
Block 1: Conrad Henry [Otto Otto Otto] Henry
Block 2: Conrad Henry [Henry Rudolf Hermann Conrad] Henry
Block 3: Conrad Henry [Frederick Henry Philip Otto Frederick] Henry
Block 4: Conrad Henry [William Richard Alfonso Rudolf Adolf Albert] Henry
```

Every block has the same outer form:

```
Conrad, Henry + x additional names + Henry
```

The value of **x** increases regularly:

```
3, 4, 5, 6
```

Formally, the rule is:

```
x(j) = 3 + j
```

Here **j** is the block number counted from zero. The first block has **j = 0** and therefore **x = 3**; the fourth has **j = 3** and therefore **x = 6**.

1.2 The historical period hypothesis

The question is not merely whether some conspicuous sequence can be found somewhere in the long history of the Empire. The hypothesis is narrower:

> The visible name structure is concentrated in a medieval period extending from 911 to 1313 and does not continue with comparable strength in the subsequent period to the end of the Empire.

The same rules are therefore applied separately to two periods:

- early study period: 911-1313;
- later comparison period: 1314-1806.

The boundary refers to the year of first royal elevation. Henry VII was elevated in 1308 and died in 1313. The next elevations in the list occurred in 1314. The year 1806 marks the dissolution of the Empire; the last ruler included, Francis II, had been elevated in 1792.

1.3 The second observation: three intervals of 113 years

A temporal regularity accompanies the name pattern. The relevant values are the death years of the regular predecessors of Conrad I through IV:

Conrad	Predecessor	Year of death
Conrad I	Louis IV the Child	911
Conrad II	Henry II	1024
Conrad III	Lothair III	1137
Conrad IV	Frederick II	1250

The differences are:

```
1024 - 911  = 113
1137 - 1024 = 113
1250 - 1137 = 113
```

The years therefore form the arithmetic progression **911, 1024, 1137, 1250**.

This second feature is not independent of the name pattern, because both involve the same four Conrads. A joint permutation test is consequently required for a joint assessment.

2. Data basis

2.1 Source and reproducibility

The ruler list was produced from the German-language Wikipedia page “Liste der römisch-deutschen Herrscher”. The stored import used for the analysis records:

- Wikipedia revision ID: **271353392**;
- retrieval time: 28 October 2026, 06:04:20 UTC;
- SHA-256 of the name file: **9ff7e9b5b1b6540ec4935b941137d9ff28a9cb0964ea1aac9153a8bca3ab30b0**.

The complete list begins in 843, after the Treaty of Verdun, and extends to the end of the Empire in 1806. It contains 64 persons.

2.2 Inclusion rules

The following rules govern the name list:

1. Entries are ordered by first royal elevation.
2. Each person appears only once.
3. A second reign or later recognition does not create another entry.
4. Kings, co-kings, and anti-kings are included.
5. Where only the year is known and a tie occurs, the order of the Wikipedia table is retained.
6. Given names are standardized for the pattern analysis. For example, **Henry II** and **Henry VII** both count as **Henry**.
7. Full names, roles, elevation years, and source references remain in the CSV audit file.

This separation is essential: the statistical analysis uses short standardized names, while historical checking continues to refer to the identified persons.

2.3 Size of the periods

Range	Number of persons	Use
843-910	6	context and complete event sequence
911-1313	31	primary name-pattern test
1314-1806	27	later comparison period
843-1806 overall	64	complete ruler list

The periods cover different numbers of calendar years but similar numbers of ruler positions. For the name patterns, the statistically relevant quantity is the order of persons, not the duration of their reigns.

2.4 Event table for the 113-year analysis

The death-year analysis uses a separate controlled table containing 52 regular predecessor-successor assignments from 843 to 1806. Each event records:

- the identity of the predecessor;
- the predecessor's year of death;
- a stable person identifier for the successor;
- the standardized given name of the successor.

Persons who remained anti-kings exclusively are not recorded as regular successors. A former anti-king who later became a regular king may be recorded in that later capacity, as in the case of Conrad III. Conrad IV had already been co-king; Frederick II is nevertheless treated as his predecessor in the regular assignment. The statistical rule uses the documented regular predecessor relation Frederick II -> Conrad IV.

3. Which patterns does the program search for?

3.1 Blocks and chains

A block consists of an opening motif, a number of filler positions, and a closing name. Several non-overlapping blocks form a chain when the number of filler positions follows a simple arithmetic rule:

```
x(j) = x0 + j x d
```

- **x0**: number of fillers in the first block;
- **j**: block number;
- **d**: step size.

The reference analysis admits **d = 0** and **d = 1**. With **d = 0**, all blocks have the same internal gap; with **d = 1**, the gap grows by one from block to block. At least three blocks are required. The reported reference configuration tolerates no deviations, omitted chain elements, or gaps that are wrong by one.

3.2 The FIXED pattern family

In **FIXED**, the opening motif and closing name remain the same in every block:

```
X, Y ... Z
X, Y .... Z
X, Y ..... Z
```

The observed pattern is **Conrad, Henry ... Henry**. The discovery search does not know these names in advance. It tests all one- and two-name opening motifs and all closing names that occur often enough to form at least three blocks. This includes forms such as **X ... X**, **X ... Y**, **X, Y ... Y**, and **X, Y ... Z**.

3.3 The BOOKEND pattern family

In **BOOKEND**, each individual block begins and ends with the same name, but the name may change between blocks:

```
A [filler names] A
B [filler names] B
C [filler names] C
```

A hypothetical constant-gap example would be **Charles [four names] Charles**, **Louis [four names] Louis**, and **Otto [four names] Otto**. The family tests an abstract bracketing structure rather than repetition of one particular name. It is included in full even though the observed winner belongs to **FIXED**.

3.4 Discovery rather than retrospective fixation

Two different questions must be distinguished:

1. How rare is the specifically fixed motif **Conrad, Henry ... Henry**?
2. How rare is the best pattern when names, pattern family, subsection, and gap rule are all selected by the program?

The second question is decisive for the principal result. Every choice allowed on the historical data is therefore allowed again in every random permutation.

4. Scoring the name patterns

4.1 Coverage

Coverage counts the positions contained within blocks. The observed pattern covers 30 of 31 positions:

```
Coverage = 30 / 31 = 0.9677 = 96.77%.
```

Only one position lies outside the four blocks.

4.2 Local density and scan score

A very short perfect hit should not automatically outrank a long, nearly closed structure. The scan score therefore combines length and density:

```
Scan score = covered positions
             x local coverage
             x deviation factor.
```

No deviations are admitted or observed, so the deviation factor is 1. For the observed pattern:

```
Scan score = 30 x (30 / 31)
           = 29.032258...
           = 29.0323.
```

For comparison across periods of different length, this value is divided by period length:

```
standardized strength = 29.0323 / 31
                      = 0.936524.
```

4.3 Why the earlier value 0.0015 is a different test

An earlier analysis scored the motif by name rarity in bits and obtained **p = 0.0015** from 2,000 permutations. That measure rewards the fourfold occurrence of a relatively rare two-name motif.

The present principal analysis instead asks how much of the list is ordered by any admissible structure. With the same list, seed, and 2,000 permutations, the coverage question gives **p = 0.0155**; with 10,000 permutations the estimate stabilizes at **p = 0.0160**.

Value	Question
p = 0.0015	How rare is an equally or more unusual name motif?
p = 0.0160	How rare is any equally or more powerful list structure?

The general pattern search therefore uses **0.0160** as its principal value.

5. The Monte Carlo permutation test

5.1 Basic idea

The observed list contains names with very different frequencies. **Henry** occurs often; other names occur once. A useful random comparison must preserve those frequencies.

The permutation test proceeds as follows:

1. All names are retained.
2. Only their order is randomly shuffled.
3. The same complete pattern search is run on the shuffled list.
4. The best random score is stored.
5. The procedure is repeated many times.

This is equivalent to shuffling a deck on which the actual ruler names are written: no cards are added or removed.

5.2 The p-value

The p-value is the proportion of random lists whose best result is at least as strong as the historical result. The plus-one correction is used:

```
p = (number of hits + 1) / (number of permutations + 1).
```

The correction prevents a p-value of exactly zero. Failure to observe a random hit in a finite simulation does not prove that the theoretical probability is zero.

5.3 What a p-value does not mean

A p-value of **0.0160** does not mean that the real list has a 1.6% probability of being random, that the pattern was intentionally created with 98.4% probability, or that a particular historical cause has been proved. It means that approximately 1.6% of randomized name orders produce an equal or higher score under the same comprehensive search.

6. Result for 911-1313

6.1 Pattern found

The complete discovery over **FIXED** and **BOOKEND** again selects:

```
Conrad Henry ... Henry
x = 3 + j.
```

Block	Index range	x	Block length
1	0-5	3	6 names
2	6-12	4	7 names
3	14-21	5	8 names
4	22-30	6	9 names

The block lengths consequently increase from 6 through 7 and 8 to 9 names.

6.2 Statistical result

With 10,000 permutations and seed 42:

```
observed scan score: 29.0323
p-value:             0.0160
significance level:  0.05
result:              significant
```

The mean of the best random scores is 18.2494 and the 95th percentile is 26.0357. The historical value of 29.0323 lies well above the typical random range, although individual random lists reach values as high as 30.

The p-value already accounts for free choice of names, one- or two-name opening motifs, closing name, **FIXED** or **BOOKEND**, **x0**, **d = 0** or **d = 1**, the best subinterval, and a minimum of three blocks.

7. Result for 1314-1806

Exactly the same search is applied to the 27 persons in the later period. Under the specified requirements, the program finds no chain of at least three blocks.

This result is not **p = 1** and does not prove the absence of every conceivable pattern. It states precisely that no structure in the implemented **FIXED** and **BOOKEND** families simultaneously meets the three-block minimum, the permitted step sizes, and zero deviation tolerance during 1314-1806. Other, unimplemented patterns may exist.

The later period therefore functions as a negative historical control: the procedure that finds a nearly complete structure in the early period finds no admissible chain under the same rules in the later period.

8. Direct comparison of the two periods

8.1 Why two separate significance decisions are insufficient

“Significant early” and “not significant late” do not automatically imply a significant difference. The difference in standardized pattern strength is therefore tested directly:

```
Difference = strength(911-1313) - strength(1314-1806)
           = 0.936524 - 0
           = 0.936524.
```

8.2 Null model for the period comparison

The 31 early and 27 late names are pooled and randomly reassigned to new groups of the same sizes. The complete **FIXED**/**BOOKEND** discovery is run in both artificial periods. This is repeated 10,000 times, counting how often the randomized difference is at least as large as the historical difference.

8.3 Result

```
one-sided comparison p-value: 0.0009
question:                     early stronger than late
result:                       significant
```

The test supports both a strong early pattern and its temporal concentration. The null model jointly captures differences in name order and name frequencies between periods. It does not separate the contribution of arrangement from historical change in the name repertoire. Both are relevant to the hypothesis of a period-specific overall structure; causal analysis would require separate modeling.

9. The 113-year progression as a separate result

9.1 General event rule

The rule is defined generally rather than tailored to four individual cases: the death year of the regular predecessor of a king is used when the king, in that regular reign, is not listed exclusively as an anti-king.

Across all 52 events, the program tests whether at least four death years form an exact arithmetic progression and whether the same successor name occupies those positions. It may search freely over successor name, starting year, ending year, and positive interval.

9.2 Uniqueness of the year progression

The 52 positions yield **C(52,4) = 270725** four-position selections. Only one consists of four different years with exactly equal spacing: **911, 1024, 1137, 1250**. The successor name at all four positions is **Conrad**.

9.3 Targeted exact p-value

If both Conrad and the four year positions are treated as fixed in advance, the four Conrad labels must occupy exactly those four positions:

```
p = 1 / C(52,4)
  = 1 / 270725
  = 0.0000036938.
```

This is about 3.7 favorable cases per million equally likely position selections.

9.4 Exact p-value corrected globally for name choice

Because Conrad was noticed from the historical data, the targeted value is too narrow for the principal assessment. Every other sufficiently frequent name must be allowed to occupy the unique four-year progression.

If a name occurs **k** times, it has **C(k,4)** ways to supply four labels. Summing over all names gives 42 favorable allocations:

```
p_global = 42 / 270725
         = 0.0001551390
         = 0.00015514.
```

This accounts for free name choice. Starting year, ending year, and interval are also globally searched because all 270,725 four-position selections were examined and only one exact year progression exists.

9.5 Scope of this null model

The test fixes the actual death years and permutes successor names. It asks how unusual it is for the unique exact four-term progression among the observed predecessor death years to carry the same successor name four times. It does not model the biological, political, or genealogical probability that these rulers died in these particular years. Such a question would require a historical-biological model of life span, elevation, co-rule, and succession.

10. Joint p-value for both features

10.1 Why the individual values cannot simply be multiplied

Multiplying **0.0160 x 0.00015514** would require suitable independence assumptions. They do not hold because both results involve the same four Conrads. A random order favoring one result may also favor the other.

10.2 Joint null model

The joint test uses one million permutations of the 31 names from 911-1313. Each permutation must meet both conditions:

1. Complete **FIXED**/**BOOKEND** discovery reaches at least the observed scan score of 29.0323.
2. The four positions of the 113-year event progression carry the same successor name.

Only permutations meeting both conditions count as hits.

10.3 Result

Ninety-six of one million random orders meet both conditions. With the plus-one correction:

```
p_joint = (96 + 1) / (1,000,000 + 1)
        = 0.0000969999
        = 0.000097.
```

This corresponds to about one joint random hit in 10,309 permutation experiments. The value is simulated rather than combinatorially exact. The 95% Wilson interval for the underlying hit rate is approximately **0.000079 to 0.000117**; even its upper bound is far below 0.05.

11. Results in context

Level	Null hypothesis	Result
Early name chain	Random order of the same 31 names; complete pattern search	**p = 0.0160**
Later period	Same search rules on 27 later names	no admissible three-block chain
Period difference	Random reassignment of 58 names to groups of 31 and 27	**p = 0.0009**
113 years, targeted	Four Conrad labels randomly allocated to 52 events	exact **p = 0.00000369**
113 years, global	Any successor name may occupy the four-term progression	exact **p = 0.00015514**
Joint result	Both structures must be at least as strong in the same permutation	**p = 0.000097**

The values answer different questions. The joint p-value does not replace the individual analyses: it combines two specified dependent features, while the period comparison tests the temporal concentration of the implemented patterns.

12. Interpretation

12.1 Results supported by the tests

Under the specified rules, the results support the following statements:

1. The name order from 911 to 1313 is unusually strong when measured by the best freely selectable **FIXED**/**BOOKEND** pattern.
2. No corresponding admissible chain is found from 1314 to 1806.
3. Standardized pattern strength is significantly concentrated in the early period.
4. The predecessor death years of the four regular Conrads form a unique exact four-term progression among all recorded events.
5. The simultaneous occurrence of the name chain and death-year progression is very rare under the joint permutation model.

The supplied replication package contains the complete 52-row event table, and every assignment is historically documented.

Reproducibility: from source retrieval to p-value

The Open Science design covers the verifiable chain from source retrieval to p-value. Every transition can be checked and the results recomputed. The starting point is the publicly accessible ruler table. Revision ID, retrieval time, and SHA-256 checksum fix the exact data state. The documented inclusion rules are then applied: first royal elevation determines order, each person appears once, kings, co-kings, and anti-kings are included, and only the given name is standardized for pattern analysis. Full names, roles, years, and sources remain in the audit data.

The replication package supplied with the article contains the unchanged name file, a CSV audit file with all 64 persons, the 52-row event table, a machine-readable configuration file, and the extensively documented analysis program. The program records the random seeds, number of permutations, admitted pattern families, minimum block count, step sizes, tolerances, score definition, and plus-one correction. A verification run also

outputs intermediate results and file checksums.

The workflow consists of five controlled stages:

1. Retrieval and local storage of the specified Wikipedia revision.
2. Selection and ordering of the 64 persons under the published rules.
3. Validation of generated files against checksums and published tables.
4. Execution of the supplied documented program with the stated seeds.
5. Comparison of scores, hit counts, p-values, and Monte Carlo intervals with the reported values.

Conclusion

The succession of rulers in the period 911-1313 shows a statistically significant name structure under the implemented search rules. The same search finds no admissible three-block chain from 1314 to 1806, and the direct period comparison confirms significant concentration in the earlier period. In addition, the death years of the regular predecessors of Conrad I-IV form the only exact four-term arithmetic progression among 52 controlled events. The joint permutation test for both dependent features gives **p = 0.000097**.

Under the specified permutation models, the joint structure is unusual. The question of its historical cause remains open and is examined separately from the statistical finding.

Appendix A: List of rulers, 843-1806

The table reproduces the complete sequence of 64 persons used in the analysis. Entries are ordered by first royal elevation or by the first status recorded in the source table, and every person appears only once. Regular kings, co-kings, and anti-kings are included. The standardized given name is solely the analysis label; historical identity remains visible in the “Person” column.

> Comparison with the research literature. For 919-1519, the persons in the list agree with the collective volume edited by Bernd Schneidmüller and Stefan Weinfurter. In addition to regular rulers, that work includes the co-kings and anti-kings present in the list, including Rudolf, Hermann, and Conrad under Henry IV, Henry under Conrad III, Henry (VII), and the kings of the Interregnum.

Source version: German Wikipedia, “Liste der römisch-deutschen Herrscher”, revision 271353392, retrieved 28 October 2026 at 06:04:20 UTC. The source page explicitly includes anti-kings. SHA-256 of the documented analysis name file: 9ff7e9b5b1b6540ec4935b941137d9ff28a9cb0964ea1aac9153a8bca3ab30b0.

Reference: Schneidmüller, Bernd / Weinfurter, Stefan (eds.) (2003): Die deutschen Herrscher des Mittelalters. Historische Portraits von Heinrich I. bis Maximilian I. (919-1519). Munich: C.H. Beck, 624 pp., ISBN 3-406-50958-4.

No.	Given name	Person (source form)	first elevation	status
1	Louis	Ludwig II. der Deutsche	843	king
2	Carloman	Karlmann	865/876	sub-king
3	Louis	Ludwig III. der Jüngere	865/876	sub-king
4	Charles	Karl III. der Dicke	865/876	king, later emperor
5	Arnulf	Arnolf von Kärnten	887	king, later emperor
6	Louis	Ludwig IV. das Kind	900	king
7	Conrad	Konrad I.	911	king
8	Henry	Heinrich I.	919	king
9	Otto	Otto I. der Große	936	king, later emperor
10	Otto	Otto II.	961	co-king, later emperor
11	Otto	Otto III.	983	king, later emperor
12	Henry	Heinrich II.	1002	king, later emperor
13	Conrad	Konrad II.	1024	king, later emperor
14	Henry	Heinrich III.	1028	co-king, later emperor
15	Henry	Heinrich IV.	1056	king, later emperor
16	Rudolf	Rudolf von Rheinfelden	1077	anti-king
17	Hermann	Hermann von Luxemburg, Graf von Salm	1081	anti-king
18	Conrad	Konrad (III.), Sohn Heinrichs IV.	1087	co-king
19	Henry	Heinrich V.	1099	co-king, later emperor
20	Lothair	Lothar III.	1125	king, later emperor
21	Conrad	Konrad III.	1138	king
22	Henry	Heinrich (VI.), Sohn Konrads III.	1147	co-king
23	Frederick	Friedrich I. Barbarossa	1152	king, later emperor
24	Henry	Heinrich VI.	1169	co-king, later emperor
25	Philip	Philipp von Schwaben	1198	king
26	Otto	Otto IV. von Braunschweig	1198	king, later emperor
27	Frederick	Friedrich II.	1212	king, later emperor
28	Henry	Heinrich (VII.)	1220	co-king
29	Conrad	Konrad IV.	1237	co-king / king
30	Henry	Heinrich Raspe	1246	anti-king
31	William	Wilhelm von Holland	1248	anti-king, later king
32	Richard	Richard von Cornwall	1257	king
33	Alfonso	Alfons von Kastilien	1257	anti-king
34	Rudolf	Rudolf I. von Habsburg	1273	king

No.	Given name	Person (source form)	first elevation	status
35	Adolf	Adolf von Nassau	1292	king
36	Albert	Albrecht I. von Habsburg	1298	king
37	Henry	Heinrich VII.	1308	king, later emperor
38	Louis	Ludwig IV., der Bayer	1314	king, later emperor
39	Frederick	Friedrich der Schöne	1314	anti-king / king
40	Charles	Karl IV. von Luxemburg	1346	anti-king, later emperor
41	Günther	Günther von Schwarzburg	1349	anti-king
42	Wenceslaus	Wenzel von Luxemburg	1376	king
43	Rupert	Ruprecht von der Pfalz	1400	king
44	Jobst	Jobst von Mähren	1410	king
45	Sigismund	Sigismund von Luxemburg	1411	king, later emperor
46	Albert	Albrecht II.	1438	king
47	Frederick	Friedrich III.	1440	king, later emperor
48	Maximilian	Maximilian I.	1486	co-king, later emperor
49	Charles	Karl V.	1519	king, later emperor
50	Ferdinand	Ferdinand I.	1531	co-king, later emperor
51	Maximilian	Maximilian II.	1562	co-king, later emperor
52	Rudolf	Rudolf II.	1575	co-king, later emperor
53	Matthias	Matthias	1612	king and emperor
54	Ferdinand	Ferdinand II.	1619	king and emperor
55	Ferdinand	Ferdinand III.	1636	co-king, later emperor
56	Ferdinand	Ferdinand IV.	1653	co-king
57	Leopold	Leopold I.	1658	king and emperor
58	Joseph	Joseph I.	1690	co-king, later emperor
59	Charles	Karl VI.	1711	king and emperor
60	Charles	Karl VII.	1742	king and emperor
61	Francis	Franz I.	1745	king and emperor
62	Joseph	Joseph II.	1764	co-king, later emperor
63	Leopold	Leopold II.	1790	king and emperor
64	Francis	Franz II.	1792	king and emperor until 1806

Note: “First elevation” determines the reproducible ordering. For Carolingian sub-kingdoms and co-kings, this date may precede a later period of sole rule. The status labels are intentionally concise; the full reign chronologies and source references remain in the published CSV audit file.